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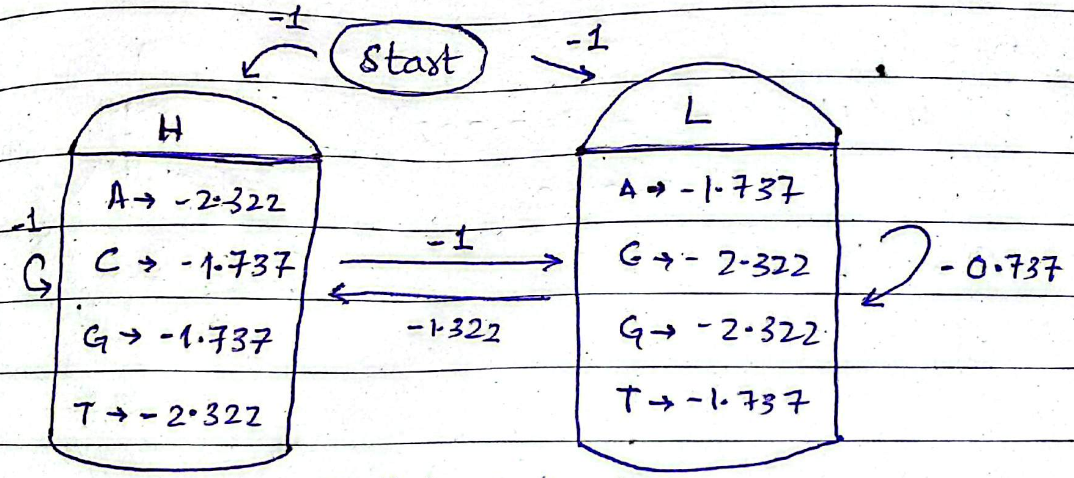
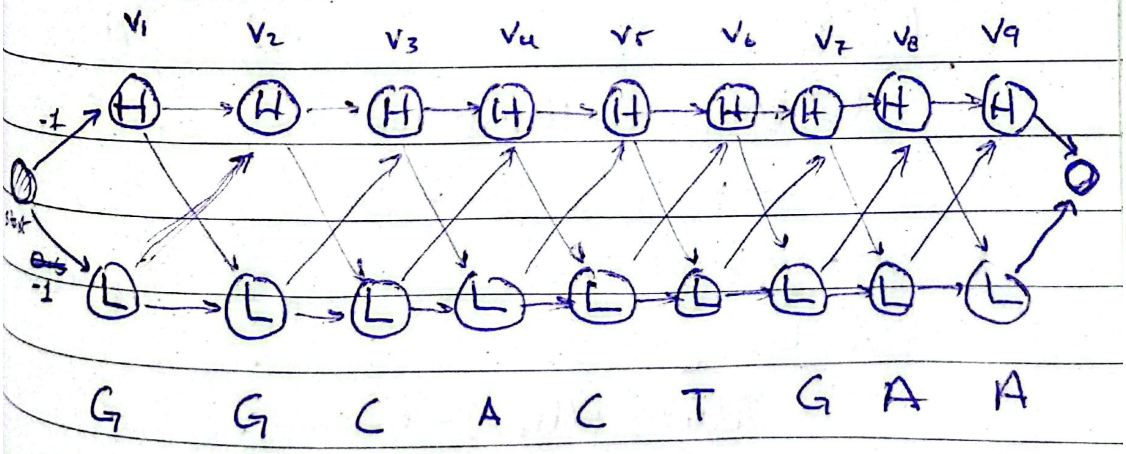
"Viterbi Algorithm"

→ Viterbi algo is used to find the best possible path of states that can generate a certain sequence of emission states.

Q: Use Viterbi Algorithm to find the most probable path of getting sequence : GGCACTGAA



→ all the probabilities are given in log



L→H

	H	L
H	-1	-1.322
L	-1	-0.737

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	A / T	G / C
H	-2.322	-1.737
L	-1.737	-2.322

- For V_1 : (G)

since we are working with logs so we will add not (x)

→ At H

$V_1(H)$ = initial prob + $P(G|H)$

$= -1 + (-1.737)$

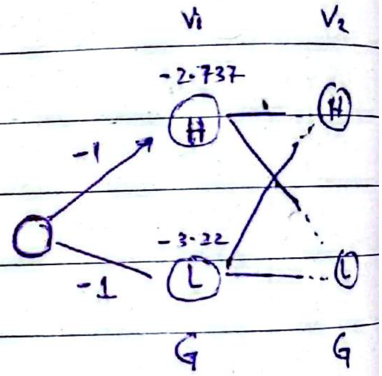
$= -2.737$

→ At L

$V_1(L) = -1 + (-2.322)$

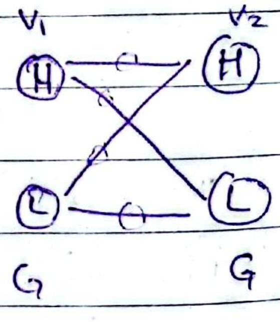
$= -3.322$

~~Choose the maximum~~



- For V_2 : (G)

we need to calc prob of each edge



→ At H:

from H:

↳ $V_1(H) + P(H|H) + P(G|H)$

$-2.737 + -1 + (-1.737)$

$= -5.474$

from L:

↳ $V_1(L) + P(H|L) + P(G|H)$

$-2.737 + (-1.322) + (-1.737)$

-6.381

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Choose the max value for $V_2(H)$.

$V_2(H) = -~~6.381~~\ 5.474 \quad (H \to H)$ with V_1 under the first H and V_2 under the second H

→ At L:

from H:

$-2.737 + P(L|H) + P(G|L)$

$-2.737 + -1 + -2.322$

$= -6.059$

from L

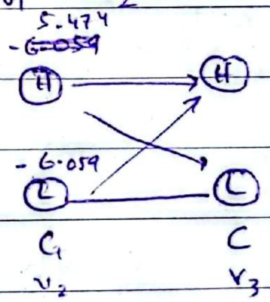
$= -3.322 + P(L|L) + P(G|L)$

$= -3.322 + (-0.737) + (-2.322)$

$= -6.381$

Choose max

$\hookrightarrow V_2(L) = -6.059 \quad (H \to L)$ with V_1 under H and V_2 under L



: For V_3 : (C)

• At H:

from H:

$= -5.474 + -1 + -1.737$

$= -8.211$

from L

$= -6.059 + (-1.322) + (-1.737)$

$= -9.118$

$V_3(H) = -8.211 \quad (H \to H)$ with V_2 under the first H and V_3 under the second H

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At L:

from H:

$= -5.474 + -1 + -2.322$

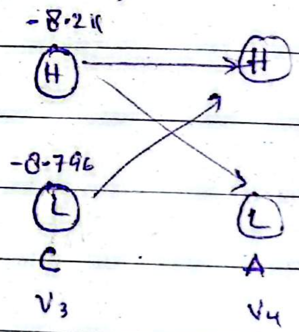
$= -8.796$

from L:

$= -6.059 + -0.737 + -2.322$

$= -9.118$

$V_3(L) = -8.796 \quad (\underset{V_2}{H} \rightarrow \underset{V_3}{L})$



V_4 : (A)

~~From~~ At H:

from ~~At~~ H:
$= -8.211 + -1 + (-2.322)$

$= -11.533$

from ~~At~~ L:

$= -8.796 + \overset{-1.322}{-9} + (-2.322$

$= -12.44$

$V_4(H) = -11.533 \quad (\underset{V_3}{H} \rightarrow \underset{V_4}{H})$

At L:

from H:

$= -8.211 + -1 + -1.737$

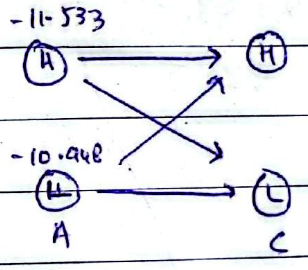
$= -10.948$

from L:

$= -8.796 + -0.737 + -1.737$

$= -11.27$

$V_4(L) = -10.948 \quad (\underset{V_3}{H} \rightarrow \underset{V_4}{L})$



• V_5 : (C)

• At H:

• from H

$= -11.533 + -1 + -1.737$

$= -14.27$

• from L

$= -10.948 + -1.322 + -1.737$

$= -14.007$ $\quad V_5(H) = -14.007$

$(\underset{V_4}{L} \rightarrow \underset{V_5}{H})$

• At L:

• from H:

$= -11.533 + -\cancel{1.322} + -\cancel{1.737}\ 2.322$

$= -14.\cancel{595}\,855$

• from L:

$= -10.948 + -0.737 + -2.322$

$= -14.007$ $\quad V_5(L) = -14.007$

$(\underset{V_4}{L} \rightarrow \underset{V_5}{L})$

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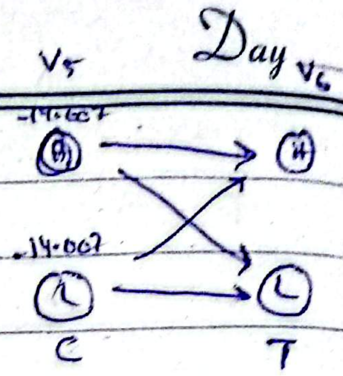


V_6 : (T)

At H:

- from H:

$= -14.007 + -1 + -2.322$

$= -17.329$

- from L

$= -14.007 + -1.322 + -2.322$

$= -17.651$

$V_6(H) = -17.329 \quad (\underset{V_5}{H} \rightarrow \underset{V_6}{H})$

At L:

- from H:

$= -14.007 + -1 + -1.737$

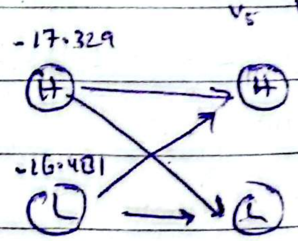
$= -16.744$

- from L:

$= -14.007 + -0.737 + -1.737$

$= -16.481$

$V_6(L) = -16.481 \quad (\underset{V_5}{L} \rightarrow \underset{V_6}{L})$



- V_7 : (G)
  - At H:
    - from H:

$= -17.329 - 1 - 1.737$

$= -20.066$

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- from L:

$= -16.481 - 1.322 - 1.737$

$= -19.54$

$V_7(H) = -19.54$ $(\underset{V_6}{L} \rightarrow \underset{V_7}{H})$

At L:

- from H:

$= -\cancel{16.481}\ 17.327 - 1 - 2.322$

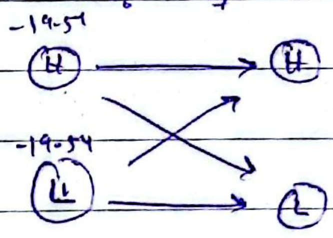
$= -20.\cancel{103}651$

- from L:

$= -16.481 - 0.737 - 2.322$

$= -19.54$

$V_7(L) = -19.54$ $(\underset{V_6}{L} \rightarrow \underset{V_7}{L})$



- V_8 : (A)
  - At H
    - from H

$= -19.54 - 1 - 2.322$

$= -22.862$

- from L

$= -19.54 - 1.322 - 2.322$

$= -23.184$

$V_8(A) = -22.862$

$(\underset{V_7}{H} \rightarrow \underset{V_8}{H})$

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- At L:
  - from H:

    $= -19.54 - 1 - 1.737$

    $= -22.277$

  - from L:

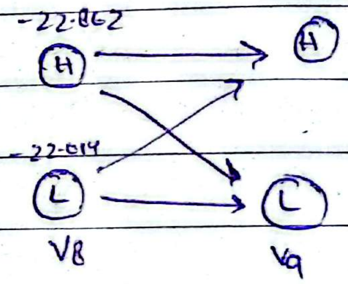
    $= -\cancel{8}19.54 - 0.737 - 1.737$

    $= -22.014$

$V_8(L) = -22.014$

$(L \to L)$ (V_7 V_8)

- V_9 : (A)



  - At H:
    - from H:

      $= -22.862 - 1 - 2.322$

      $= -26.184$

    - from L:

      $= -22.014 - 1.322 - 2.322$

      $= -25.658$

$V_9(H) = -25.658$

$(L \to H)$ (V_8 V_9)

  - At L:
    - from H:

      $= -22.862 - 1 - 1.737$

      $= -25.599$

$V_9(L) = -24.488$

$(L \to L)$ (V_8 V_9)

    - from L:

      $= -22.014 - 0.737 - 1.737$

      $= -24.488$

$V_3 \leftarrow V_a$

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↳ Now choose Maximum

$$\max\left(V_9(H),\ V_9(L)\right)$$

$$\max\left(-25.658,\ -24.488\right)$$

↳ max → L

↳ Backtracking from $V_9(L)$:

$$V_9(L) \leftrightarrows V_8(L) \leftarrow V_7(L) \leftarrow V_6(L) \leftarrow V_5(L)$$

$$V_1(H) \longrightarrow V_2(H) \longrightarrow V_3(H) \rightarrow V_4(L)$$

↳ $\boxed{\text{Path : HHHLLLLLL}}$